Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf ____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _S0/0/0___________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _129________

Does R2 show up as an OSPF neighbor on R1? __Yes_______

Does R2 show up as an OSPF neighbor on R3? __No_______

What does this information tell you?

From R3 to the 192.168.2.0/24 network, this traffic will be routed through R1. As S0/0/1 is configured as a passive interface, OSPF information is not being advertised here. The 129 accumulated cost metric for the 192.168.2.0/24 network on R3 is 64+64+1, which corresponds to two T1 serial links (cost 64) and the R2 Gigabit 0/0 LAN link (cost 1). ______________________________

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1_______________________________________

R2(config-router)# no passive-interface s0/0/1_______________________________

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65, this is from 64+1, corresponds to T1 serial links (cost 64) and R2 Gigabit 0/0 LAN link (cost 1)__

Is R2 listed as an OSPF neighbor to R3? _Yes________

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The OSPF default reference-bandwidth is 100 Mb/s, since modern link speeds are faster than that. To calculate the cost accurately, the default reference-bandwidth setting needs to be changed to a higher value.

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Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 = 782 = 781+1 = R1: S0/0/1 (cost 781) + R3: G0/0 (cost 1)

The cost to 192.168.23.0/30 = 845 = 781+64 = R1: S0/0/1 (cost 781) + R3: S0/0/1 (cost 64)

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562 = 781+781. Before R3: S0/0/1 with cost 64; now it is R3: S0/0/1 with cost 781, so the new accumulated cost is R1: S0/0/1 (cost 781) + R3: S0/0/1 (cost 781).

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because now the route to the 192.168.3.0/24 network on R1 go with R1: S0/0/0 (cost 781) + R2: S0/0/1 (cost 781) + R3: G0/0 (cost 1) = 1563, which is lower than the route go through R3 (the route that not go through R2) of R1: S0/0/1 (cost 1565) + R3: G0/0 (cost 1) =1566. OSPF will choose the route with the least cost, so the route is now going through R2.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   This is because the router ID affects the designated router (DR) and backup designated router (BDR) election: it first checks the priority; if the priority is the same, it checks the router ID. If the active interface is set and that interface goes down, the router ID may change. This causes OSPF to reset neighbor relationships and trigger a new DR/BDR election. To prevent the unstable situation, controlling the router ID assignment is important, such as manually setting the router ID or using a loopback interface.

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2. Why is the DR/BDR election process not a concern in this lab?

   This is because in this lab, routers are connected with point-to-point serial links. Since the DR/BDR election occurs only on multi-access networks, no DR/BDR election is required.

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3. Why would you want to set an OSPF interface to passive?

   Setting an OSPF interface to passive allows it to advertise the network to its neighbors without sending Hello packets. This can reduce unnecessary OSPF traffic and save bandwidth.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   Using OSPF areas can divide a network into smaller sections. By making the protocol hierarchical, OSPF can reduce routing traffic and calculations within each area, so it can support scalable networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used for the routing decision. Since routers compare Administrative Distance first, EIGRP has the lowest administrative distance (90), which is preferred over OSPF (110) and RIP (120).

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